

**MASTER OF COMPUTER APPLICATION/
MASTER OF SCIENCE (INFORMATION TECHNOLOGY)**
ACADEMIC YEAR:2026-2027(ODD) SEMESTER: 3

PAPER CODE: 101300317

PAPER TITLE: PRACTICAL BASED ON ADVANCED DATA STRUCTURE

SR. NO.	PRACTICAL LIST
1)	Write a program to check whether a number is even or odd using function.
2)	Write a program to swap two numbers using a temporary variable and without using a third variable.
3)	Write a program to find the largest among three numbers using ternary operator.
4)	Write a program to find the smallest among four numbers using else if statement.
5)	Write a program to print the Fibonacci series up to n terms using functions.
6)	Write a program to check whether a year is a leap year.
7)	Write a program to calculate factorial of a number using a function
8)	Write a program to check whether a string is a palindrome.
9)	Write a program to display the multiplication table.
10)	Write a program to implement a calculator using switch statement (with function).
11)	Write a program to find the number is Armstrong or not.
12)	Write a program to implement a function to add two numbers.
13)	Write a program to demonstrate call by value and call by reference.
14)	Write a program to reverse a string without using library functions.
15)	Write a program to count the number of vowels, consonants, digits, and spaces in a string.
16)	Write a program to print the Fibonacci series up to n terms using functions.
17)	Write a program to calculate the factorial of a number using recursion.
18)	Write a program to implement function overloading (e.g., area of square, rectangle, and circle).
19)	Write a program to create a class Student with data members and member functions.
20)	Write a program to implement constructor and destructor in a class.
21)	Write a program to demonstrate the use of static data members and static member functions.
22)	Write a program to demonstrate Single Inheritance.
23)	Write a program to demonstrate Multilevel Inheritance.
24)	Write a program to demonstrate Multiple Inheritance.

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25)	Write a program to swap two numbers using pointers.
26)	Write a program to implement virtual functions and runtime polymorphism.
27)	Write a program to demonstrate the use of Enumeration.
28)	Write a program to demonstrate the use of array of pointers.
29)	Write a program to demonstrate the Operator Overloading.
30)	Write a program to demonstrate the Friend Class .
31)	1) Write a program to insert/delete in a linear array at specific position. 2) Write a program to remove duplicate elements from linear array. 3) Write a program to read 10 integers in an array. Sort them in ascending order.
32)	1) Write a program to implement stack using array that performs the following operations. (a) PUSH, (b) POP, (c) PEEK, (d) ISEMPY, (e) ISFULL, (f) DISPLAY 2) Write a program to reverse array elements using stack.
33)	1) Write a program to implement queue using array that performs the following operations (a) ENQUEUE, (b) DEQUEUE, (c) ISEMPY, (d) ISFULL, (e) DISPLAY 2) Write a program to implement circular queue using array that performs the following operations (a) ENQUEUE, (b) DEQUEUE, (c) ISEMPY, (d) ISFULL, (e) DISPLAY
34)	Write a menu driven program to implement singly linked list that performs the following operations (a) createList(): To create the list with 'n' number of nodes initially as defined by the user. (b) traverse(): The given traverse() function traverses and prints the content of the linked list. (c) insertAtFront(): This function simply inserts an element at the front/beginning of the linked list.

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35)	Write a menu driven program to implement following functions in a singly linked list (a) insertAtEnd(): This function inserts an element at the end of the linked list. (b) insertAtPosition(): This function inserts an element at a specified position in the linked list. (c) deleteFirst(): This function simply deletes an element from the front/beginning of the linked list. (d) deleteEnd(): This function simply deletes an element from the end of the linked list. (e) deletePosition(): This function deletes an element from a specified position in the linked list. (f) reverseLL(): This function reverses the given linked list.
36)	1) Write a program to construct binary tree using array. 2) Write a program to traverse binary search tree using linked list.
37)	1) Write a program to implement depth-first search. 2) Write a program to implement breadth-first search.
38)	Write a program to implement linear search and binary search algorithms.
39)	Write a program to implement insertion sort and selection sort algorithms.
40)	Write a program to implement quick sort and merge sort algorithms.